An X-linked additive kernel for the Eftedal et al. relative categories

Derivation and analysis for `eftedal_reanalysis` (Phase 2)

2026-10-20

Question. The Eftedal et al. (2025) correlations split each of 16 relationship types by family side (maternal, paternal, other) and by the sexes of the pair (ssf, ssm, os), giving 82 categories. Same-sex pairs correlate slightly more than opposite-sex pairs, and maternal relatives more than paternal ones. The Phase-2 models need a parameter-free genetic route for such differences. The X chromosome is the obvious candidate: its inheritance depends on the sexes along the pedigree path. What is its kernel for each category, how does assortative mating (AM) change it, and how much of the observed differences can it explain?

Answer.

1. **Kernel.** With full dosage compensation (FDC), the X covariance between relatives is proportional to the X kinship ϕ^X. On the scale of the autosomal kernel, $K^X = \frac{4}{3}\phi^X$. All 82 categories are tabulated, averaging over sexes the data do not record (Table 4).
2. **Size.** X should carry about $s_X \approx 2\%$ of V_A (plausible range 1–5%), not its 5% share of the genome, because under FDC an X locus contributes 3/8 of an autosomal locus's variance.
3. **Assortative mating** does not simply scale the X kernel by the autosomal AM factor; that approximation errs by up to 0.75 in K^X. AM's excess sharing comes from mate correlation on the *total* genetic value, and X takes part in it roughly additively, with sex-specific transmission. An exact sex-aware path model gives the correction $D_k(m) = K_k^X(m) - K_k^{\text{auto}}(m)$ on a grid of the mate correlation m.
4. **Signatures.** At $s_X = 0.02$, X explains $\leq 7\%$ of the sibling same-sex excess and of the maternal half-sibling excess, with the wrong sex pattern for the latter. It explains about 15–35% of the small maternal excess among 1st and 2nd cousins, and most of it only at the top of the plausible range. It explains $\leq 3\%$ of the cousins' same-sex excess. Other routes, such as maternal-line or sex-specific environment, are needed.

Code and tables are in `theory/x_kernel/` (Section 8).

1 The categories and what "side" means

Eftedal et al. (2025, Fig. 1 legend) define side through the connecting pair one generation above the relatives. Two women make the pair maternal, two men paternal, and a woman and a man "other". Full siblings are "other" because they are connected through both parents. Concretely:

type	maternal	paternal	other
half sibs	shared mother	shared father	–
1st cousins	parents are sisters	parents are brothers	brother and sister
1st cousins t/uz	parents are twin sisters	twin brothers	–
half cousins	parents are half sisters	half brothers	one of each
2nd cousins (and t/uz)	parents are two female 1st cousins	two male 1st cousins	one of each
co-cousins	connecting mates on both mothers' side	both fathers' side	mixed
step sibs	–	–	always

Several sexes are not recorded. These are averaged with equal weights:

- the sibling pair at the top of a 2nd-cousin pedigree (FF, FM, MF, MM);
- the shared grandparent of half cousins;
- the twins' sex for 2nd cousins t/uz;
- which member of an os pair is female (this matters only for asymmetric pedigrees).

Same-sex twins of unknown zygosity are $\frac{1}{2}$ MZ, $\frac{1}{2}$ DZ, as in our model. Every configuration and its weight is in `x_configs.csv`. Moving any of these weights by 10 percentage points changes K^X by at most 0.02, i.e. ≤ 0.0003 in correlation. No side definition was ambiguous.

2 Random mating

X kinship. ϕ^X_{ij} is the probability that an X allele drawn at random from i and one from j are identical by descent; a male has one X. The standard recursion runs on the younger member i (who is not an ancestor of j):

$$\phi^X_{ij} = \begin{cases} \phi^X_{\mathrm{mo}(i),\,j} & i \text{ male} \\ \frac{1}{2}\big(\phi^X_{\mathrm{mo}(i),\,j} + \phi^X_{\mathrm{fa}(i),\,j}\big) & i \text{ female} \end{cases}, \qquad \phi^X_{ii} = \begin{cases} 1 & \text{male} \\ \frac{1}{2} & \text{female} \end{cases},$$

with unrelated founders at 0. For example, sisters $\frac{3}{8}$, brothers $\frac{1}{2}$, brother–sister $\frac{1}{4}$. Maternal half siblings are $\frac{1}{8}$ (sisters), $\frac{1}{2}$ (brothers) and $\frac{1}{4}$ (os). Paternal half sisters are $\frac{1}{4}$, and any paternal half-sibling pair with a male is 0.

Genetic values and dosage compensation. Consider one X locus with allele frequency p, centred allele values u ($\mathrm{Var}\, u = pq$), and effect b per *active* copy.

- *FDC* (random X inactivation): a male's value is $b\,u$. A female is a mosaic with value $\frac{b}{2}(u_1{+}u_2)$. The variances are b^2pq (male) and $b^2pq/2$ (female).
- *No dosage compensation (NDC)*: a female's value is $b(u_1 + u_2)$, so the variances are b^2pq (male) and $2b^2pq$ (female).

The covariance between relatives is $\sum_{u\in i,\, v\in j} w_u w_v\, pq\, \Pr(u \equiv v)$, with allele weights w. Under FDC ($w = b$ for males, $b/2$ for females) every sex combination gives $\mathrm{Cov}[X_i, X_j] = b^2pq\,\phi^X_{ij}$. Under NDC ($w = b$ throughout) it gives $4b^2pq\,\phi^X_{ij}\,2^{-\#\mathrm{males}}$.

We put X on the scale of the autosomal kernel by dividing by the *sex-averaged* X variance $\bar{V}_X$:

$$\mathrm{Cov}[X_i, X_j] = \bar{V}_X \, K^X_{ij}, \qquad K^X_{\mathrm{FDC}} = \tfrac{4}{3}\,\phi^X, \qquad K^X_{\mathrm{NDC}} = \tfrac{4}{1.5}\,\phi^X\, 2^{-\#\mathrm{males}}.$$

K^X_{ii} averages to 1 over the sexes. MZ females are $\frac{2}{3}$ and MZ males $\frac{4}{3}$ under FDC, which says only that X variance is twice as large in males. Table 4 gives the kernel for every biological category. Step siblings, co-cousins and adoptive pairs are 0 at random mating.

3 How much of V_A is X-linked

- **FDC is supported.** Sidorenko et al. (2019) estimate a male/female ratio of X-linked h^2_{SNP} of 1.9–2.2 across 20 UK Biobank traits (FDC predicts 2, NDC $\frac{1}{2}$).
- **Per-length expectation.** X is 5.0% of the haploid genome (156 of about 3,100 Mb), and about 4% of protein-coding genes. An autosomal locus with per-allele effect b contributes $2b^2pq$. Under FDC an X locus with the same per-active-copy effect contributes b^2pq in males and $b^2pq/2$ in females, so 3/8 of an autosomal locus on average. Hence

$$s_X = \frac{0.05 \times \frac{3}{8}}{0.95 + 0.05 \times \frac{3}{8}} \approx 0.019 \qquad (\mathrm{NDC}:\ \approx 0.038).$$

- **Empirical check.** In UK Biobank, X explains 0.63% (males) and 0.30% (females) of phenotypic variance on average. That is roughly 1–3% of the traits' total h^2_{SNP}, and lower per Mb than the autosomes, as haploidy predicts (Sidorenko et al. 2019).
- **Recommendation.** Fix $s_X = 0.02$, with plausible range 0.01–0.05. The upper end allows for enrichment of brain and intellectual-disability genes on X and for partial escape from X inactivation.

4 Assortative mating

Sex-aware model. Each person carries an autosomal value A^a and an X value X, with $A = A^a + X$ and $Y = A + E$. Under FDC, writing V_{Xf0} for the base female X variance (the male one is $2V_{Xf0}$):

$$\begin{aligned} A^a_o &= \tfrac{1}{2}\big(A^a_{\mathrm{mo}} + A^a_{\mathrm{fa}}\big) + e^a, & \mathrm{Var}\, e^a &= \tfrac{1}{2}V_{a0}, \\ X_{\mathrm{daughter}} &= \tfrac{1}{2}X_{\mathrm{mo}} + \tfrac{1}{2}X_{\mathrm{fa}} + e_d, & \mathrm{Var}\, e_d &= \tfrac{1}{4}V_{Xf0}, \\ X_{\mathrm{son}} &= X_{\mathrm{mo}} + e_s, & \mathrm{Var}\, e_s &= V_{Xf0}. \end{aligned}$$

A son's X value is his mother's transmitted allele counted at full male dosage, so it regresses on her value with coefficient 1 and on his father's with 0. Segregation variances keep their random-mating values, as in Fisher's treatment. Mates are matched on Y with correlation ρ_y (Gaussian assortment), so for the vector $z = (A^a, X)$

$$\mathrm{Cov}[z_{\mathrm{mo}}, z_{\mathrm{fa}}] = \rho_y \, \Sigma_F \mathbf{1}\mathbf{1}^\top \Sigma_M / (\sigma_{YF}\sigma_{YM}),$$

and the per-sex covariance matrices Σ_F, Σ_M are iterated to equilibrium. The genotypic mate correlation is $m = \mathrm{Corr}[A_{\mathrm{mo}}, A_{\mathrm{fa}}]$. For each category the stationary model is built in pathMgr over its pedigree configurations, with one co-path per union. This includes the childless unions that connect step siblings and co-cousins.

Definition used by the model. At matched m, run the model with an X share s and without ($s = 0$, giving $K_k^{\text{auto}}(m)$), and define K^X by the decomposition the Phase-2 model uses:

$$K_k^{\text{total}}(m) = (1-s)\,K_k^{\text{auto}}(m) + s\,K_k^X(m) \quad \Longleftrightarrow \quad K_k^{\text{total}}(m) = K_k^{\text{auto}}(m) + s\,D_k(m),$$

with $D_k = K_k^X - K_k^{\text{auto}}$. K^X absorbs the A–X cross covariances that AM creates; they are first order in s.

Checks.

- At $m = 0$ the model reproduces the closed forms of Section 2 for all 82 categories (max error $< 10^{-6}$).
- The autosomal part reproduces Fisher's sibling kernel $(1 + m)/2$ (max error $< 10^{-6}$).
- K^X does not depend on s: opposite-sex siblings at $m = 0.43$ give 0.5899, 0.5901 and 0.5903 at $s = 0.005, 0.02, 0.05$.
- At fixed m, changing h^2 from 0.5 to 0.78 moves D by at most 0.034, i.e. ≤ 0.0005 in correlation.

Results. Figure 1 and Table 2 show $D_k(m)$.

1. **Siblings: AM shrinks the correction.** From $m = 0$ to $m = 0.43$, D goes from $+0.17$ to $+0.04$ for brothers and from -0.17 to -0.13 for opposite-sex pairs, because AM-induced sharing is common to X and autosomes.
2. **AM widens the side contrast for distant relatives.** Paternal-side cousin pairs that include a male cannot share X through the father–son link, yet the autosomal excess grows with m, so their D becomes more negative. Maternal-side pairs keep their advantage, because mother-to-son X transmission has coefficient 1. The maternal$-$paternal contrast in D grows from 0.24 to 0.34 for 1st cousins and from 0.06 to 0.18 for 2nd cousins.
3. **In-laws acquire small corrections**, e.g. $D = -0.11$ for paternal co-cousins, both male. Their autosomal kernel is pure AM, and X reaches them through the same sex-specific transmission.

Approximations. Table 3 compares two shortcuts with the exact K^X.

- *Autosomal AM factor on the random-mating X kernel*, $K^X(0)\,K^{\text{auto}}(m)/K^{\text{auto}}(0)$, is poor. It treats X's AM excess as proportional to its random-mating sharing, so it gives zero excess to every pair with $\phi^X = 0$, which is wrong.
- *A constant correction*, $D_k(m) \approx D_k(0) = K_{\text{FDC}}^X - a_k$, errs by ≤ 0.002 in correlation at $s_X = 0.02$, $V_A = 0.5$. It is acceptable if interpolation is inconvenient.

5 GxG, mtDNA and Y

- **GxG.** No X version of the a^2 kernel is needed. Pairs involving X carry about $2s_X \approx 4\%$ of GxG variance, and replacing a by the X kernel for them moves the GxG kernel by $\lesssim 0.01$, i.e. ≤ 0.002 in correlation.
- **mtDNA** acts as a maternal-line indicator (column `mt` of Table 4): 1 for siblings, maternal half siblings and maternal 1st cousins; $\frac{1}{2}$ for maternal half cousins; $\frac{1}{4}$ for maternal 2nd cousins. Its variance for cognitive traits is expected to be negligible, and the indicator is confounded with any maternal-line environment.
- **Y** is shared only by paternal-line male pairs; it is ignored (`y_share` in `x_kinship.csv`).

6 Predicted signatures against the data

X adds $V_A\, s_X\, D_k(m)$ to the correlation of category k. Contrasts are formed identically for the data and for the prediction:

- same-sex minus opposite-sex: per side, $\frac{1}{2}(\text{ssf} + \text{ssm}) - \text{os}$, then averaged over sides;
- maternal minus paternal: per sex composition, then averaged.

Observed standard errors assume independent category estimates. Table 1 and Figure 2 use $m = 0.43$.

1. **Siblings and half siblings: X is far too small.** It accounts for $\leq 7\%$ of the sibling same-sex excess (0.030) and of the maternal half-sibling excess (0.050), and $\leq 18\%$ even at $s_X = 0.05$. The sex pattern also disagrees. X puts the maternal half-sibling excess in brothers ($+0.005$) and opposite-sex pairs ($+0.003$), with sisters slightly *paternal* (-0.0005). The data show the excess in all three compositions (ssf $+0.035$, ssm $+0.062$, os $+0.052$).
2. **Cousins' maternal excess: X can contribute.** It accounts for about 15–35% of the 1st- and 2nd-cousin excess (0.014 and 0.0095) at central values, and most of it only at $s_X = 0.05$, $V_A = 0.7$.
3. **Cousins' same-sex excess: X is negligible** ($\leq 3\%$).
4. **Half cousins** show a paternal excess (-0.013 ± 0.011), the opposite of X's prediction, but the observed value is within about 1.5 SE of it.
5. **No dosage compensation** changes the predicted contrasts by at most a factor of 2.5 and leaves every conclusion unchanged (`signatures.csv`).

7 Implications for the Phase-2 model

1. Use $K_k^{\text{total}}(m) = K_k^{\text{auto}}(m) + s_X\, D_k(m)$. Keep our own K^{auto} (Nagylaki half siblings, corrected step siblings, and so on), and interpolate D_k linearly in m from `x_kernel_grid.csv`.

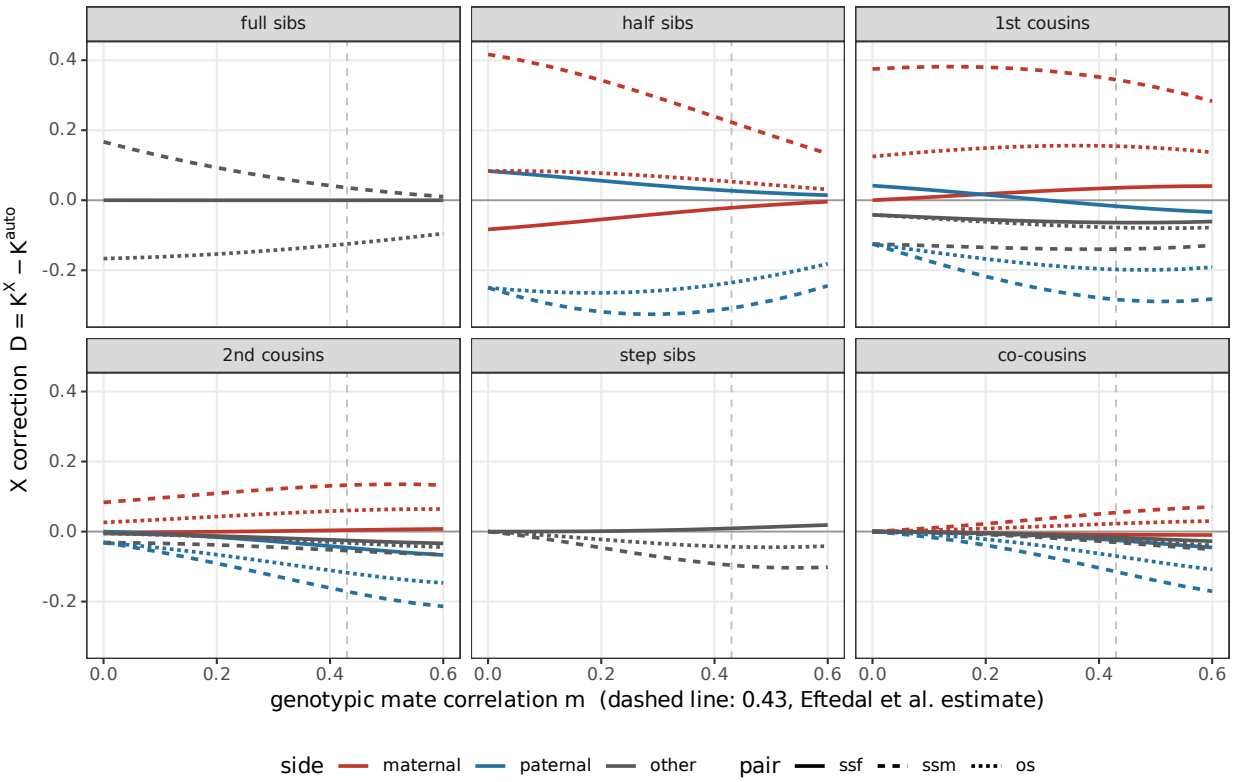


Figure 1: The X correction $D = K^X - K^{\text{auto}}$ against the genotypic mate correlation m (exact, FDC). The effect on a correlation is $V_A\, s_X\, D$: about $0.01\, D$ at $s_X = 0.02$, $V_A = 0.5$.

2. Fix $s_X = 0.02$, and report sensitivity at 0.05. Estimating s_X freely would mainly absorb the environmental side and sex effects that X cannot produce.

3. Expect X to change fitted sex and side multipliers only slightly. Its largest per-category effect is ≤ 0.005 in correlation.

Caveats.

- The model assumes primary phenotypic AM with Gaussian assortment at equilibrium, unlinked loci, no escape from X inactivation and no sex-specific autosomal effects.
- The 2nd cousins / other / ssf row is a suspected duplicate (`data/README.md`). It enters only the 2nd-cousin same-sex contrast.
- The equal sex weights could be replaced by registry frequencies, but the effect is negligible (Section 1).

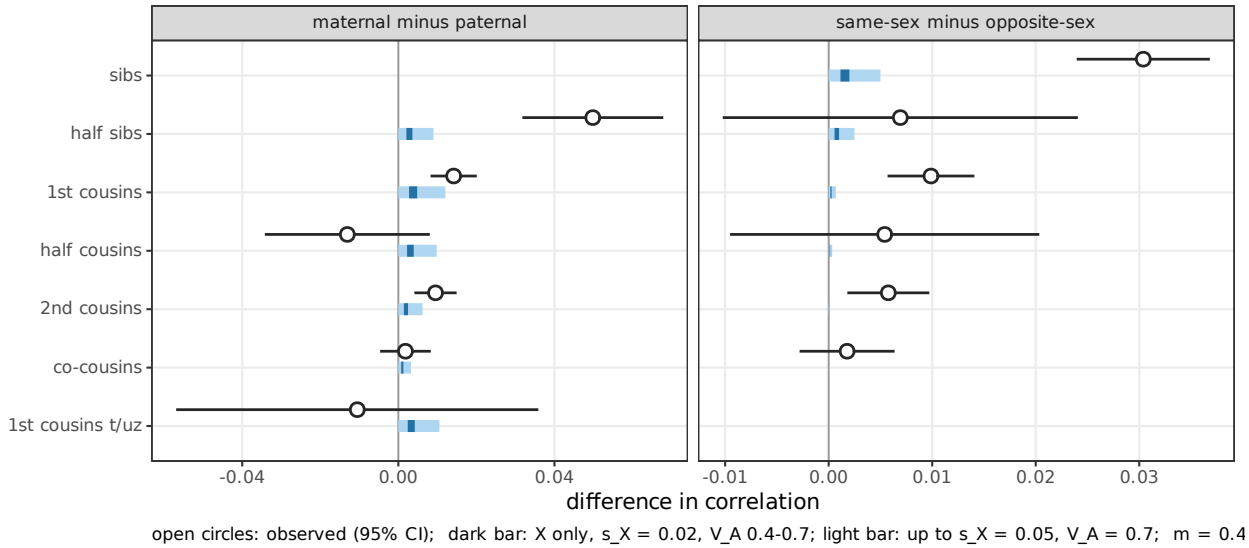


Figure 2: Observed contrasts (open circles, 95% CI) against X alone (bars), at $m = 0.43$.

Table 1: Predicted X-only contrasts against the data. "Explained" is the prediction at $s_X = 0.02$, $V_A = 0.7$ as a share of the observed contrast, shown where the observed contrast is beyond 2 SE.

		observed		X only, $m = 0.43$		
contrast	type	estimate	SE	s_X=0.02, V_A 0.4–0.7	s_X=0.05, V_A=0.7	explained
ss − os	dz	-0.0305	0.0342	+0.0011 – +0.0020	+0.0050	n.s.
ss − os	sibs	+0.0304	0.0033	+0.0011 – +0.0020	+0.0050	7%
ss − os	half sibs	+0.0069	0.0088	+0.0006 – +0.0010	+0.0025	n.s.
ss − os	1st cousins	+0.0099	0.0021	+0.0002 – +0.0003	+0.0007	3%
ss − os	half cousins	+0.0054	0.0076	+0.0001 – +0.0001	+0.0003	n.s.
ss − os	2nd cousins	+0.0058	0.0020	0.0000 – +0.0001	+0.0001	<1%
ss − os	step sibs	+0.0137	0.0197	0.0000 – 0.0000	0.0000	n.s.
ss − os	cocousins	+0.0018	0.0023	0.0000 – 0.0000	0.0000	n.s.
mat − pat	half sibs	+0.0498	0.0092	+0.0021 – +0.0036	+0.0090	7%
mat − pat	1st cousins	+0.0142	0.0030	+0.0027 – +0.0048	+0.0120	34%
mat − pat	half cousins	-0.0131	0.0108	+0.0022 – +0.0039	+0.0098	n.s.
mat − pat	2nd cousins	+0.0095	0.0028	+0.0014 – +0.0025	+0.0062	26%
mat − pat	1st cousins t/uz	-0.0105	0.0237	+0.0024 – +0.0042	+0.0105	n.s.
mat − pat	cocousins	+0.0018	0.0033	+0.0007 – +0.0013	+0.0032	n.s.

Table 2: Exact K^X and D under AM for selected categories (FDC).

		$m = 0$		$m = 0.2$		$m = 0.43$	
category	sex	K^X	D	K^X	D	K^X	D
sibs, other	ssf	0.500	0.000	0.600	0.000	0.715	0.000
sibs, other	ssm	0.667	0.167	0.693	0.093	0.751	0.036
sibs, other	os	0.333	-0.167	0.446	-0.154	0.590	-0.125
half sibs, maternal	ssf	0.167	-0.083	0.313	-0.055	0.516	-0.022
half sibs, maternal	ssm	0.667	0.417	0.711	0.343	0.760	0.223
half sibs, maternal	os	0.333	0.083	0.445	0.077	0.591	0.053
half sibs, paternal	ssf	0.333	0.083	0.424	0.056	0.565	0.027
half sibs, paternal	ssm	0.000	-0.250	0.049	-0.319	0.229	-0.308
half sibs, paternal	os	0.000	-0.250	0.103	-0.265	0.302	-0.235
1st cousins, maternal	ssf	0.125	0.000	0.234	0.018	0.401	0.035
1st cousins, maternal	ssm	0.500	0.375	0.596	0.380	0.710	0.344
1st cousins, maternal	os	0.250	0.125	0.365	0.149	0.520	0.155
1st cousins, paternal	ssf	0.167	0.042	0.232	0.016	0.349	-0.017
1st cousins, paternal	ssm	0.000	-0.125	-0.002	-0.218	0.082	-0.284
1st cousins, paternal	os	0.000	-0.125	0.048	-0.168	0.168	-0.198
1st cousins, other	ssf	0.083	-0.042	0.161	-0.055	0.302	-0.064
1st cousins, other	ssm	0.000	-0.125	0.081	-0.135	0.226	-0.140
1st cousins, other	os	0.083	-0.042	0.154	-0.062	0.288	-0.078
2nd cousins, maternal	ssf	0.029	-0.003	0.077	-0.001	0.191	0.004
2nd cousins, maternal	ssm	0.115	0.083	0.187	0.109	0.319	0.133
2nd cousins, maternal	os	0.057	0.026	0.121	0.043	0.247	0.060
2nd cousins, paternal	ssf	0.031	0.000	0.062	-0.016	0.141	-0.046
2nd cousins, paternal	ssm	0.000	-0.031	-0.013	-0.091	0.016	-0.171
2nd cousins, paternal	os	0.000	-0.031	0.012	-0.066	0.069	-0.118
step sibs, other	ssf	0.000	0.000	0.094	0.001	0.311	0.009
step sibs, other	ssm	0.000	0.000	0.046	-0.046	0.206	-0.096
step sibs, other	os	0.000	0.000	0.070	-0.022	0.258	-0.043
cocousins, maternal	ssf	0.000	0.000	0.021	-0.004	0.104	-0.009
cocousins, maternal	ssm	0.000	0.000	0.048	0.022	0.167	0.054
cocousins, maternal	os	0.000	0.000	0.035	0.009	0.135	0.023
cocousins, paternal	ssf	0.000	0.000	0.021	-0.005	0.088	-0.025
cocousins, paternal	ssm	0.000	0.000	-0.013	-0.039	-0.002	-0.115
cocousins, paternal	os	0.000	0.000	0.004	-0.022	0.042	-0.070

Table 3: Approximation errors against the exact K^X over the 58 biological categories. $|\Delta\rho|$ is the error in correlation at $s_X = 0.02$, $V_A = 0.5$.

m	approximation	median $\lvert\Delta K^X\rvert$	max $\lvert\Delta K^X\rvert$	max $\lvert\Delta\rho\rvert$
0.2	constant D: $K^X(0) + K^{auto}(m) - a$	0.025	0.093	0.0009
0.2	AM factor: $K^X(0)\,K^{auto}(m)/a$	0.046	0.271	0.0027
0.43	constant D: $K^X(0) + K^{auto}(m) - a$	0.049	0.194	0.0019
0.43	AM factor: $K^X(0)\,K^{auto}(m)/a$	0.136	0.752	0.0075

Table 4: Random-mating X kernels for the biological categories. a is autosomal relatedness, FDC is $\frac{4}{3}\phi^X$, NDC the alternative convention, and mt the probability of sharing mtDNA.

		FDC (default)			NDC			
category	a	ssf	ssm	os	ssf	ssm	os	mt
mz	1.000	0.667	1.333	–	1.333	0.667	–	1.00
uz	0.750	0.583	1.000	–	1.167	0.500	–	1.00
dz	0.500	0.500	0.667	0.333	1.000	0.333	0.333	1.00
sibs	0.500	0.500	0.667	0.333	1.000	0.333	0.333	1.00
half sibs, maternal	0.250	0.167	0.667	0.333	0.333	0.333	0.333	1.00
half sibs, paternal	0.250	0.333	0.000	0.000	0.667	0.000	0.000	0.00
1st cousins t/uz, maternal	0.188	0.146	0.583	0.292	0.292	0.292	0.292	1.00
1st cousins t/uz, paternal	0.188	0.250	0.000	0.000	0.500	0.000	0.000	0.00
1st cousins, maternal	0.125	0.125	0.500	0.250	0.250	0.250	0.250	1.00
1st cousins, paternal	0.125	0.167	0.000	0.000	0.333	0.000	0.000	0.00
1st cousins, other	0.125	0.083	0.000	0.083	0.167	0.000	0.083	0.00
half cousins, maternal	0.062	0.062	0.250	0.125	0.125	0.125	0.125	0.50
half cousins, paternal	0.062	0.083	0.000	0.000	0.167	0.000	0.000	0.00
half cousins, other	0.062	0.042	0.000	0.042	0.083	0.000	0.042	0.00
2nd cousins t/uz, maternal	0.047	0.049	0.198	0.099	0.099	0.099	0.099	0.50
2nd cousins t/uz, paternal	0.047	0.073	0.000	0.000	0.146	0.000	0.000	0.00
2nd cousins t/uz, other	0.047	0.036	0.000	0.036	0.073	0.000	0.036	0.00
2nd cousins, maternal	0.031	0.029	0.115	0.057	0.057	0.057	0.057	0.25
2nd cousins, paternal	0.031	0.031	0.000	0.000	0.062	0.000	0.000	0.00
2nd cousins, other	0.031	0.026	0.000	0.026	0.052	0.000	0.026	0.00

8 Files

file	content
`pedigrees.py`	pedigree configurations per (rel, side, sex), with weights
`x_kernel.py`	X kinship, random-mating kernels, sex-aware AM equilibrium and pathMgr model
`compute.py`	`x_kinship.csv`, `x_configs.csv`
`grid.py`	`x_kernel_grid.csv`: K^{auto}, K^{X}, D for $m = 0, 0.05, \ldots, 0.6$ and 0.43
`signatures.py`	`signatures.csv`: predicted and observed contrasts
`make_tables.py`, `checks.py`, `figure.R`	tables, checks and figures for this report
`x_kernel.md`	two-page summary

Reproduce: `PYTHONPATH=<worktree>/src python compute.py && python grid.py && python signatures.py && python make_tables.py && python checks.py && Rscript figure.R`, then `pdflatex x_kernel.tex` twice.

References. Eftedal et al. (2025), *PNAS*, PMC12207517. Sidorenko et al. (2019), *Nat. Commun.* 10:3009, PMC6614401. Yang et al. (2011), *Nat. Genet.* 43:519–525. Fernando & Grossman (1990), *Theor. Appl. Genet.* 80:75–80. Fisher (1918), *Trans. R. Soc. Edinb.* 52:399–433.